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Enhanced methane production and energy potential from rice straw by employing microaerobic pretreatment *via* anaerobic digestion

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ABSTRACT

This study employs a microaerobic pretreatment that includes different microbial reagents combined with oxygen to enhance the energy conversion potential of rice straw (RS) *via* anaerobic digestion (AD) in a more cost-effective and time-efficient manner. Five broad varieties of microbial reagents including liquid consortium (LC), cow manure (CM), sheep dung (SD), biogas slurry (BS), and straw-decomposing consortia (SC), have been used as sources of useful microbes at varying oxygen concentrations of 0, 6, 12 and 24 mL_{O2}/gVS to explore the effect of microaerobic pretreatment on RS degradation. The results showed that RS pretreated with SC had the highest cumulative methane yield (CMY) of 311.7 mL/gVS at 12 mL_{O2}/gVS, followed by BS, CM, LC, and SD, with the CMYs of 270.5, 263.2, 257.1, and 256.7 mL/gVS at 6 mL_{O2}/gVS, respectively. The RS pretreated with SC, BS, CM, LC, and SD led to improvements in the methane yield by 88.7%, 63.7%, 59.3%, 55.6%, and 55.4%, respectively, compared to the untreated RS. The microbial community analysis revealed that the relative abundance (RA) of *Firmicutes*, which are known to play a key role in accelerating the rate of hydrolysis, increased after microbial pretreatment combined with oxygen supply. The RA of *Clostridium III*, which facilitates cell wall degradation *via* hydrolytic pathways along with other sugar-fermenting bacteria, such as *Saccharofermentans*, *Lutaonella*, and *Sedimentibacter,* increased after AD in the digestate of pretreated RS compared to that in untreated RS. Moreover, a shift in acetoclastic and hydrogenotrophic archaea, including *Methanothrix* and *Methanospirillum,* was also observed, indicating precursor changes resulted by pretreatments and varied microbial metabolism, which might have enhanced the methane production process of RS. The energy potential analysis revealed that microbial pretreatment was a suitable technique for stimulating the energy conversion efficiency of RS. This study can be used as a reference for future utilization of a diverse range of lignocellulosic materials for clean energy production.

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1. Introduction

Rapid economic development and increasing population have led to a growing energy demand, with global energy consumption reaching 12.93 billion tons in 2014 (Mustafa et al., 2016). Nonrenewable energy in the form of fossil fuels accounted for 86.3% of the total energy consumption worldwide, whereas renewable energy accounted for only 9.2%. The high dependency on fossil fuels has resulted in increased greenhouse gas (GHG) emissions and thus contributed to climate change, eventually causing adverse environmental impacts. Hence, there is a dire need to develop environmentally friendly and cost-effective alternative energy sources to meet the growing energy demand, one of which is biomass energy (Li et al., 2019). Different types of feedstocks, including agricultural waste, municipal solid waste (MSW), industrial waste, *etc*., are used as biomass energy sources to produce biomethane *via* anaerobic digestion (AD), which is a cost-effective technology adopted for clean energy production (Jingura and Kamusoko, 2017; Xin et al., 2020).

Rice straw (RS) is a lignocellulosic biomass that is abundantly generated in China, with a total yield of 213 million tons in 2017 (Xia et al., 2018). RS consists of a complex matrix of cellulose, hemicellulose, and lignin, which makes it more recalcitrant to biodegradation (Huang et al., 2020). Therefore, several pretreatment methods have been studied for removing these barriers and

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List of abbreviations

AD	Anaerobic Digestion
ADF	Acid Detergent Fiber
ADL	Acid Detergent Lignin
ANOVA	Analysis of Variance
BD	Biodegradability
B_0	Simulated Maximum Cumulative Methane Yield
BS	Biogas Slurry
CM	Cow Manure
CMY	Cumulative Methane Yield
CrI	Crystallinity Index
d	Day
DMY	Daily Methane Yield
DNA	Deoxyribonucleic Acid
EMY	Experimental Methane Yield
FTIR	Fourier Transform Infrared Spectroscopy
GC	Gas Chromatograph
GHG	Greenhouse Gas
h	Hour
LC	Liquid Consortium
NDF	Neutral Detergent Fiber
OTU	Operational Taxonomic Unit
R^2	Correlation Coefficient
RA	Relative Abundance
RS	Rice Straw
SC	Straw-decomposing Consortia
SD	Sheep Dung
SEM	Scanning Electron Microscope
TA	Total Alkalinity
TAN	Total Ammonia Nitrogen
TMY	Theoretical Methane Yield
TS	Total Solids
URS	Untreated Rice Straw
VFAs	Volatile Fatty Acids
VS	Volatile Solids
XRD	X-Ray Diffraction
λ	Lag Phase Time
μ_m	Maximum Methane Production Rate

enhancing the enzymatic hydrolysis of RS to improve its energy output in the form of biomethane.

To date, physical, chemical, and biological pretreatments have been widely applied to pretreat RS (Guan et al., 2018). Physical methods are applied to reduce the particle size of the substrate, whereas chemical pretreatment involves the use of chemicals to enhance biomass digestibility; however, these pretreatments have high energy costs and require careful handling (Harmsen et al., 2010; Li et al., 2015). Biological pretreatment is an environmentally friendly and cost-effective method that employs fungi to degrade lignocellulosic components *via* the action of various useful enzymes (Saritha et al., 2012). The use of fungi reduces pretreatment cost because these organisms are capable of degrading lignin with the help of ligninolytic enzymes; however, the degradation rate is slow. Hence, a longer pretreatment time and higher oxygen consumption are required during fungal pretreatment (Kamusoko et al., 2019). Therefore, it is important to find an effective method with a shorter pretreatment time for faster hydrolysis of lignocellulosic components. As reported in a previous study, microaerobic pretreatment is a time-efficient method that requires a shorter pretreatment time than other methods. During this pretreatment process, microorganisms are partially exposed to oxygen under moderate operating parameters. The process accelerates the hydrolysis stage by stimulating microbial activity and cell growth (Botheju and Bakke, 2011; Fu et al., 2015; Montalvo et al., 2016; Wagner et al., 2018; Xu et al., 2018). Although oxygen may help regulate the microbial community structure, it is still challenging to balance the positive and negative effects of microaeration during AD. Moreover, detailed analysis of the mechanisms involved in the microaerobic pretreatment is also required to further understand the process.

To carry out the microaerobic pretreatment, this study chose different types of microbial reagents that contain complex mixtures of useful microorganisms belonging to different growth environments and are known to play a key role in hydrolyzing cellulose. These microorganism-rich reagents, including liquid consortium (LC), cow manure (CM), sheep dung (SD), biogas slurry (BS), and straw-decomposing consortia (SC), were used as a potential source of different microbes to investigate the microbial pretreatment effects on the AD performance of RS. This study proposed an economical microaerobic pretreatment method for the efficient degradation of lignocellulosic substrates with reduced pretreatment time while enhanced methane production. It also established the connection between the oxygen supply and microbial functions and how this interaction impacts the rate of hydrolysis.

The main objectives of this study are: (1) developing a sustainable pretreatment method by using different microbial reagents and oxygen levels to enhance the methane production performance of RS, (2) investigating the changes in the microbial communities before pretreatment, after pretreatment, and after AD to study the response of functional microorganisms to varying oxygen concentrations, and (3) carrying out energy potential analyses that consider different microbial reagents and their respective biomethane yields.

2. Materials and methods

2.1. Feedstock and inoculum

RS was collected from a farm located in Henan Province of China, and it was chopped to a size of less than 1 cm and ground by a high-speed grinder (XINGSHILIHE, Beijing, China). Sheep dung (SD) and cow manure (CM) were collected from farms in Shunyi District, Beijing, China, and deionized water was added to each manure to make a homogenous mixture. Liquid consortium (LC) and straw-decomposing consortia (SC) are commercially available and were purchased for this experiment. Biogas slurry (BS) used as the startup inoculum for AD was obtained from the Beijing Donghuashan Biogas Plant.

2.2. Analytical methods

Total solids (TS) and volatile solids (VS) of the feedstock and inoculum were analyzed as per standard methods (APHA, 2005). The substrates were weighed and placed in an oven at 105 °C for 24 h to remove moisture and weighed again. The TS content was calculated from the weight difference, as shown in Eq. (1). The dried samples were placed in a muffle furnace at 550 °C for 2 h, and then they were removed, placed in a desiccator, cooled to room temperature, and weighed. The VS content was calculated as shown in Eq. (2).

$$TS = \frac{m_3 - m_1}{m_2 - m_1} \times 100\% \quad (1)$$

$$VS = \frac{m_3 - m_4}{m_2 - m_1} \times 100\% \quad (2)$$

where m_1 indicates the weight of the empty aluminum pan, m_2 refers to the weight of the aluminum pan and substrate before drying, m_3 refers to the weight of the aluminum pan and substrate after drying, and m_4 is the weight of the aluminum pan and ash after placing in the muffle furnace. The elemental C, H, N, and S compositions were quantified by an elemental analyzer (Vario EL cube, Elementar, Germany). The O content of RS was determined by assuming C + H + O + N = 99.5% on a VS basis (Zhang et al., 2018). The pH was measured with a pH meter (Mettler Toledo, USA). The analysis of cellulose, hemicellulose, and lignin was performed according to the Van Soest method (Soest et al., 1991) using an A2000 fiber analyzer (ANKOM, USA) by determining the neutral detergent fiber (NDF), acid detergent fiber (ADF), and acid detergent lignin (ADL) contents, respectively (Li et al., 2018b). The calculation of daily biogas production was performed based on the difference in the pressure measured by a barometer (WAL Mess-und Regelsysteme GmbH, Germany) before and after releasing biogas from each AD digester. The biogas yield was adjusted to the standard condition (0 °C, 101 kPa) with the ideal gas law (Li et al., 2013), as shown in Eq. (3).

$$V_{biogas} = \frac{P \times V_{headspace} \times C}{R \times T} \quad (3)$$

where V_{biogas} represents the daily biogas yield (L), P represents difference in pressure (kPa) before and after releasing the biogas, $V_{headspace}$ represents the volume of the headspace (L) in the anaerobic digester, C represents the molar volume (22.41 L/mol), T represents the absolute temperature (K), and R represents the universal gas constant (8.314 L kPa/K/mol).

The composition of biogas was determined using a gas chromatograph (GC) system (7890B, Agilent, USA) equipped with an analytical column (Agilent Hayesep Q) and a thermal conductivity detector. Operation temperatures of 60 and 220 °C were set for the column oven and detector, respectively, and the carrier gas was helium. Volatile fatty acids (VFAs) were quantified by a GC system (7890A, Agilent, USA) equipped with a DB-WAXETR capillary column (30 m × 530 μm × 1.0 μm) and a flame ionization detector, and their operation temperatures were 50 and 250 °C, respectively. The concentrations of total ammonia nitrogen (TAN) and total alkalinity (TA) were measured using HACH test kits (HACH, USA) according to the manufacturer's instructions. The structural features of untreated and pretreated RS were measured by a Nicolet 6700 Fourier transform infrared spectrophotometer (FTIR) (Thermo Fisher Scientific, Waltham, MA) in the range of 400−4000 cm^{-1}. Untreated and pretreated samples were analyzed by grinding with KBr (1:100, w/w) and pressing into transparent pellets. Moreover, the morphology of the treated and untreated samples was investigated by scanning electron microscope (SEM, Hitachi S-4700, Japan) at a magnification of 500. The crystallinity index (*CrI*) of the cellulose component, which was obtained before and after pretreatment of the samples, was observed by X-ray diffraction (XRD) profiles. Bruker D8-Advance was used at 40 kV and 40 mA with Cu Kα radiation to examine the XRD patterns. Scans were obtained from 5 to 60° at a rate of 5°/min. The *CrI* of the pretreated RS was calculated by Eq. (4).

$$CrI = [\ (I_{002} - I_{amorphous}) / I_{002}\ ] \times 100\% \quad (4)$$

where *CrI* refers to the crystallinity index (%), I_{002} indicates the maximal diffraction intensity of the 002-lattice plane, and $I_{amorphous}$ stands for the intensity of the background peak at $2\theta = 18°$.

2.3. Effects of microbial reagents and oxygen on the pretreatment of rice straw

Pretreatment was performed in triplicate using RS with an organic loading of 20 g VS/L. 5 mL LC and 4 g VS/L CM, SD, BS, and SC were separately used as microbial reagents to pretreat the RS. Deionized water was added to achieve a working volume of 100 mL, and the retention time was 24 h. N_2 was flushed in each bottle for 3 min to ensure the removal of air, and rubber stoppers were used to seal the bottles. Then, controlled oxygen concentrations of 0, 6, 12, and 24 mL_{O2}/gVS were injected into the AD digesters. The bottles were then placed in an incubator at 37 °C. A gas chromatograph (7890B, Agilent, USA) was used to measure oxygen and CO_2 concentration every 6 h until all oxygen was completely used.

2.4. Anaerobic digestion

AD was performed in triplicate. When all the oxygen was consumed, the concentrated inoculum was inoculated into the pretreated RS at a feedstock to inoculum ratio of 1. A working volume of 0.25 L was achieved using deionized water. The digesters were purged with high-purity N_2 to ensure anaerobic condition, after which they were sealed and placed in a 37 °C incubator for 50 days. Three blank digesters (control group) were used to subtract background CH_4 from the test group. All digesters were shaken at a speed of 130 rpm every day for 1 min before testing the biogas pressure. A schematic illustration of the experimental design is shown in Fig. 1.

2.5. Theoretical maximum methane yield (TMY) and biodegradability (BD)

TMY (mL/gVS) was calculated based on the elemental composition of feedstock by employing Buswell's and Chen's formula, as per Eqs. (5) and (6), respectively (Buswell and Mueller, 1952; Shen et al., 2019).

$$C_nH_aO_bN_c + \left(n - \frac{a}{4} - \frac{b}{2} + \frac{3c}{4}\right)H_2O \rightarrow \left(\frac{n}{2} + \frac{a}{8} - \frac{b}{4} - \frac{3c}{8}\right)CH_4 + \left(\frac{n}{2} - \frac{a}{8} + \frac{b}{4} + \frac{3c}{8}\right)CO_2 + cNH_3 \quad (5)$$

$$TMY = \frac{22.4 \times 1000 \times \left(\frac{n}{2} + \frac{a}{8} - \frac{b}{4} - \frac{3c}{8}\right)}{12n + a + 16b + 14c} \times (1 - Lignin\%) \quad (6)$$

BD was determined according to the experimental methane yield (highest cumulative methane yield obtained from experiment, EMY) and TMY, as shown in Eq. (7).

$$BD = \frac{EMY}{TMY} \times 100\% \quad (7)$$

2.6. Kinetic model

The modified Gompertz model shown in Eq. (8) was employed for the simulation and description of the AD process.

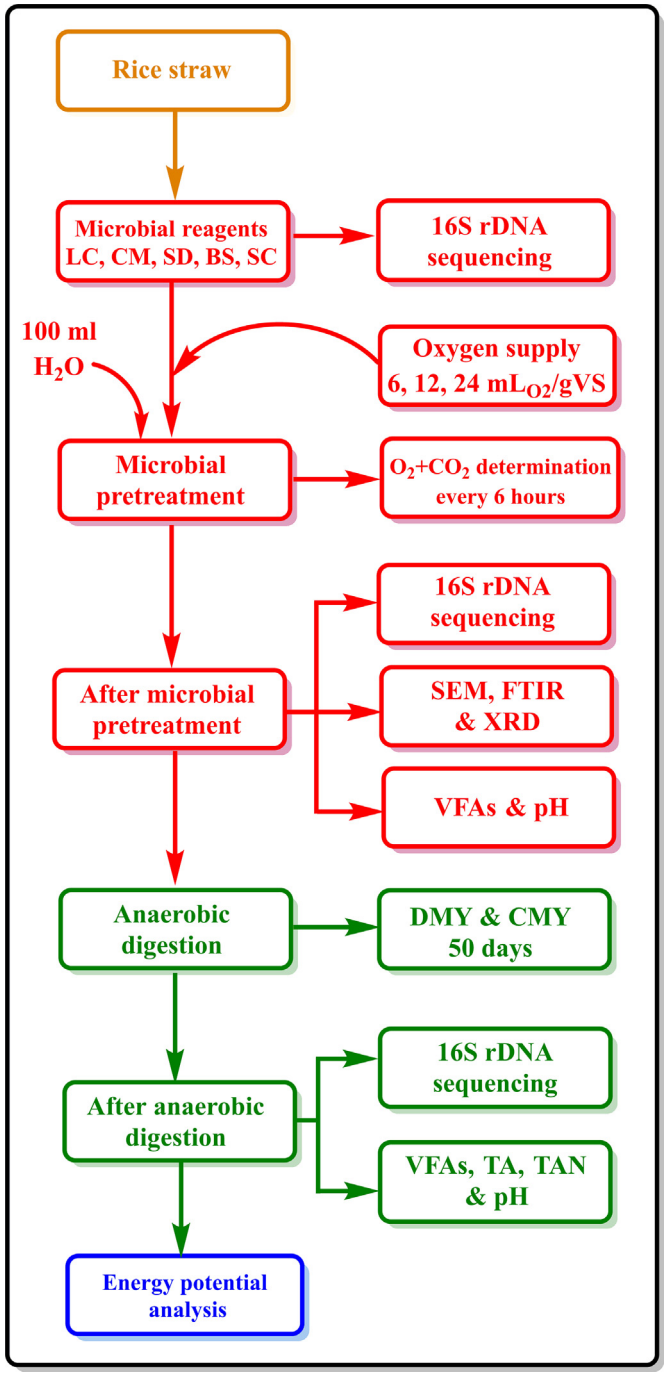


Fig. 1. Schematic diagram illustrating the experimental design.

$$B = B_0 \exp\left\{-\exp\left[\frac{\mu_m e}{B_0}\left(\lambda - t\right) + 1\right]\right\} \quad (8)$$

where B indicates the simulated cumulative methane yield (mL/gVS), B_0 represents the simulated maximum cumulative methane yield (mL/gVS), μ_m refers to the maximum methane production rate (mL/gVS/d), e is equal to 2.718, λ denotes the lag phase time (d), and t indicates the digestion time (d).

2.7. Microbial community analysis

Digestates were sampled for archaeal and bacterial genomic DNA extraction during three different stages: before pretreatment, after pretreatment, and after AD. The microbes in the digestates were characterized by employing 16S rDNA gene sequencing. For identifying bacteria, 805R (5′-GACTACHVGGGTATCTAATCC-3′) and 341F (5′-CCTACGGGNGGCWGCAG-3′) were the primer pair for amplification. The identification of archaea was carried out in two amplification round with two different primer pairs: 109F (5′-ACTGCTCAGTAACACGT-3′) and 1492R (5′-CGGCTACCTTGTTACGAC-3′) were used in the first round and 519R (5′-CGTATTACCGCGGCTGCTGG-3′) and 109F (5′-ACTGCTCAGTAACACGT-3′) were used in the second round. Cutadapt software was used for preprocessing amplicon sequences. The PEAR program was used to merge paired-end reads with a maximum error rate of 0.1. The operational taxonomic units (OTUs) were clustered and taxonomically classified with the RDP Classifier. The relative abundance (RA) was calculated by dividing the number of sequences with the total sequences affiliated with a taxon (Mao et al., 2019). The applied Shannon index for the species diversity estimation was calculated using MOTHUR tools as well as the Chao1 and ACE estimators for the richness estimation (Li et al., 2018b).

2.8. Energy potential analysis

The energy potential analysis was evaluated in the form of energy density and total energy yield (Passos et al., 2017; Shen et al., 2019).

2.9. Statistical analysis

All the data and graphs were processed using Origin Pro 9.0 (Origin Lab, USA). An analysis of variance (ANOVA) was used to statistically analyze the data.

3. Results and discussion

3.1. Characteristics of RS and inoculum

Table 1 shows the characteristics of the substrate and inoculum. The untreated rice straw (URS) had TS and VS values of 92.7% and 77.9%, respectively. The VS/TS ratio of RS was 84.0, indicating a high organic content, which is preferred for AD. The cellulose, hemicellulose, and lignin contents were 33.9%, 27.5%, and 8.0%, respectively, which was comparable with previously published values (Guan et al., 2018; Kainthola et al., 2019). The TMY of RS was 432.8 mL/gVS based on Eq. (6). The N, C, H, O, and S contents of RS were 1.1%, 40.3%, 5.6%, 37.2%, and 0.1%, respectively. The C/N ratios of the RS and inoculum were 36.6 and 9.5, respectively. The

Table 1
Characteristics of feedstock and inoculum.

Parameter	RS	Inoculum
TS (%)[a]	92.7 ± 0.1	12.0 ± 0.0
VS (%)[a]	77.9 ± 0.3	6.8 ± 0.0
VS/TS (%)	84.0	56.6
pH	ND	7.5 ± 0.7
Cellulose (%)[b]	33.9 ± 0.4	ND
Hemicellulose (%)[b]	27.5 ± 0.3	ND
Lignin (%)[b]	8.0 ± 0.4	ND
N (%)[b]	1.1 ± 0.1	2.8 ± 0.0
C (%)[b]	40.3 ± 0.1	26.6 ± 0.1
H (%)[b]	5.6 ± 0.0	4.5 ± 0.0
O (%)[b]	37.2 ± 0.1	ND
S (%)[b]	0.1 ± 0.0	1.1 ± 0.0
C/N	36.6	9.5

[a] Based on the total sample weight.
[b] On TS basis, RS: rice straw, ND: not detectable.

characteristics of the chosen substrate showed that it was a potential feedstock for producing biomethane *via* AD.

3.2. Variations in O_2 and CO_2 concentration during microbial pretreatment

Oxygen was supplied at four different concentrations of 0, 6, 12, and 24 mL_{O2}/gVS to explore its effect on the microbial community structure in the digester. Microbes play an important role in hydrolyzing polysaccharides into monosaccharides, which are partially converted into CO_2 with the supply of oxygen (Sadhu and Maiti, 2013; Shallom and Shoham, 2003). All oxygen was consumed in 24 h by different microbial reagents, namely, LC, CM, SD, BS, and SC, as shown in Fig. 2. At an oxygen concentration of 6 mL_{O2}/gVS, the fastest consumption of oxygen was observed by SC in 6 h, followed by CM and BS in 12 h, while LC and SD consumed all the oxygen in 18 h. At the oxygen concentration of 12 mL_{O2}/gVS, the fastest consumption of oxygen was again observed by SC in 6 h, followed by CM and BS, which took 18 h for oxygen consumption, while LC and SD consumed all the oxygen in 24 h. At the oxygen concentration of 24 mL_{O2}/gVS, SC consumed all the oxygen in 12 h, followed by LC, CM, SD, and BS, which took 24 h to consume all the oxygen. Accordingly, in LC, CM, SD and BS, a rise in CO_2 content was seen, which shows that microbes present in these microbial reagents had moderate metabolic activity, while the CO_2 content in SC showed rapid metabolism during the oxygen supply, thus confirming a high activity of microbial communities. Hence, the degradation of RS pretreated by different microbial reagents in the presence of oxygen can be evaluated by the production of CO_2. A previous report indicated that during pretreatment of lignocellulosic material, cellulose and hemicellulose could be depolymerized to glucose and xylose, which were further utilized by microorganisms to produce CO_2 in the aerobic environment (Lu et al., 2019; Shallom and Shoham, 2003), whereas lignin might be converted to vanillic acid by adopting different pathways and transformed into pyruvic and α-ketobutyric acid (Bugg et al., 2011).

3.3. VFAs and pH after pretreatment

The changes in VFAs and pH after pretreatment are shown in Fig. 3F. The desirable VFAs produced during pretreatment are acetate and butyrate, which contribute to methane production; however, the generation of other VFAs, such as valeric and propionic acid, is known to limit the process of methane production (Xu et al., 2014). The total VFAs after pretreatment with different microbial reagents in the absence of oxygen, such as LC, CM, SD, BS, and SC, were found to be 1345.4, 2105.7, 2140.9, 2431.4, and 2032.2 mg/L, respectively. At the oxygen concentration of 6 mL_{O2}/gVS, the highest VFAs concentrations in the pretreated groups by LC, CM, SD, and BS were 2440.0, 2506.3, 2937.8, and 2810.0 mg/L, respectively, whereas for the SC pretreated group, the concentration of VFAs was the highest (2525.9 mg/L) at 12 mL_{O2}/gVS. This finding indicated a significant increase in VFAs production, which was observed when microbial pretreatment was combined with oxygen. This might be attributed to the enrichment of *Firmicutes* and *Bacteroidetes* during pretreatment, which requires further verification by detailed study on microbial community analysis. As microbes belonging to these two phyla mostly contain hydrolytic bacteria, which might have contributed to the hydrolysis of macromolecules to produce VFAs at a faster pace, therefore, a higher VFAs concentration was achieved. Furthermore, the concentrations of acetate and butyrate were the top two highest in all groups pretreated with microbial reagents except for the SD-pretreated group, which had the highest concentration of propionic acid at 430–990 mg/L. This difference might lead to the lower methane yield of the SD-pretreated group compared to the other pretreated groups because previous studies have reported that a higher concentration of propionic acid adversely affects the function of methanogenic bacteria and thereby lowers the methane yield (Lee et al., 2019; Wang et al., 2009). At an oxygen concentration of 24 mL_{O2}/gVS during pretreatment, the concentration of total VFAs decreased, which might be due to the oxidation of readily available VFAs during oxygen pretreatment. These results are consistent with the findings of a previous study that supplied oxygen into the AD digester while treating food waste (Xu et al., 2014). The initial pH of

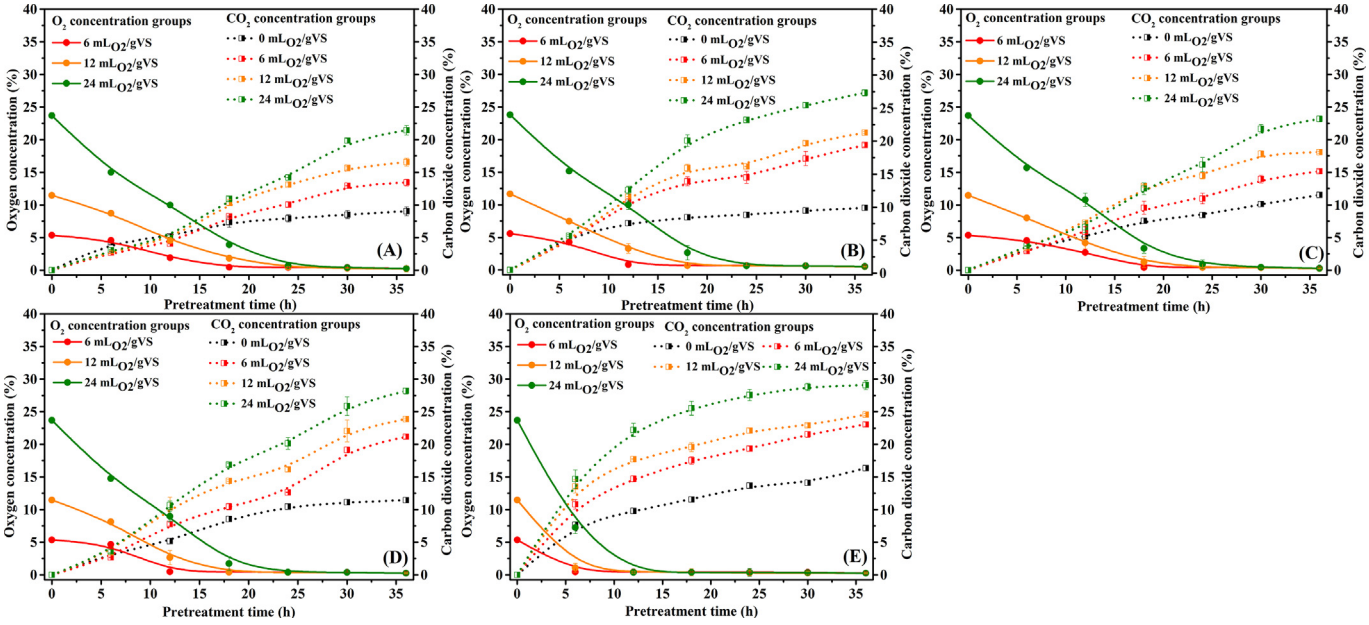


Fig. 2. Determination of O_2 and CO_2 concentrations during the pretreatment of RS with different microbial reagents: (A) LC, (B) CM, (C) SD, (D) BS, and (E) SC at different oxygen concentrations.

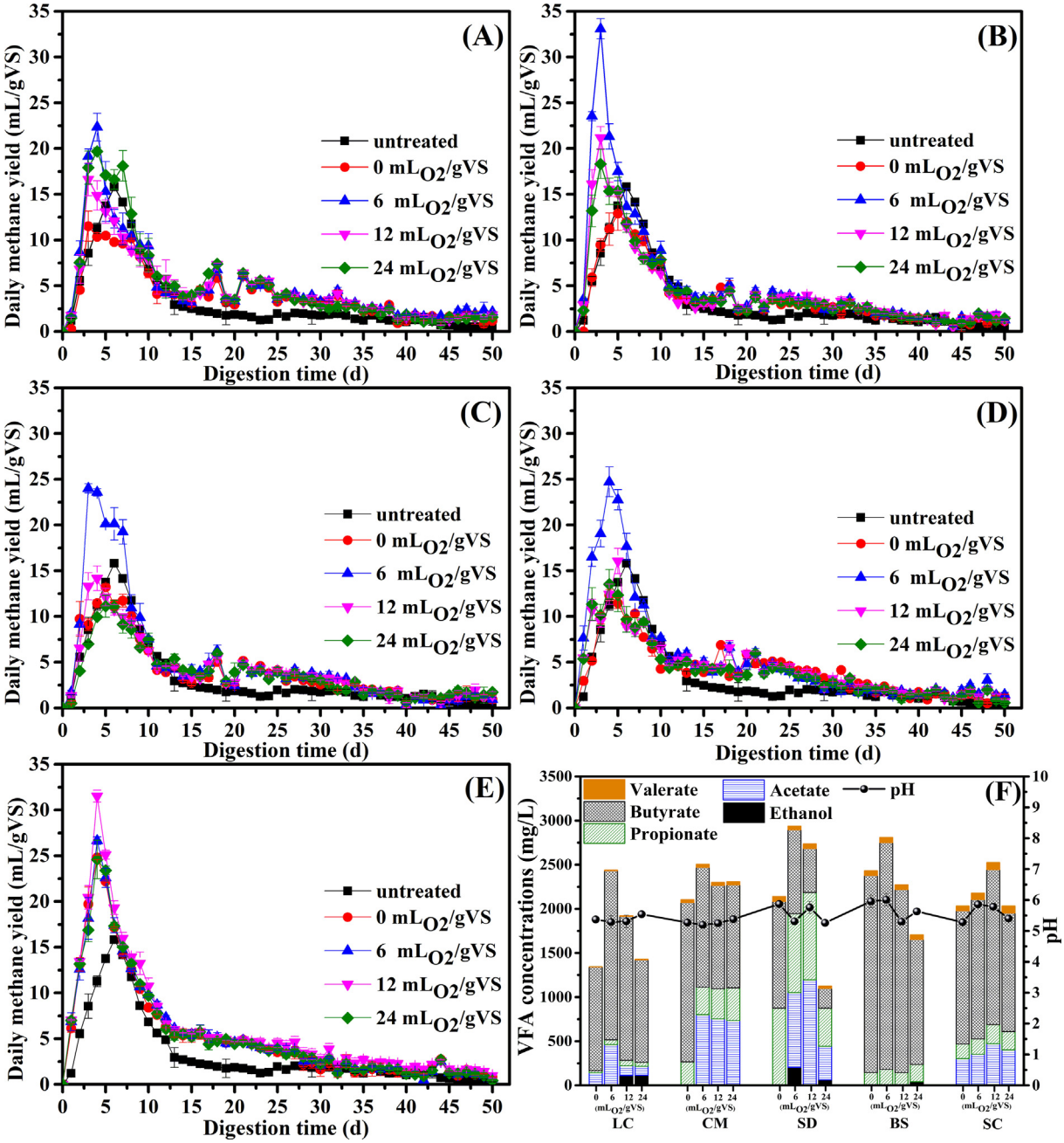


Fig. 3. Daily methane yield of RS pretreated with different microbial reagents: (A) LC, (B) CM, (C) SD, (D) BS, (E) SC, and (F) VFAs and pH levels at different oxygen concentrations.

the digester was 7.2–7.5, which dropped to 5.2–5.9 in all groups after pretreatment due to the production of VFAs.

3.4. Effect of microbial pretreatment on methane production

3.4.1. Daily methane yield (DMY)

The main purpose of microbial pretreatment is to enhance the methane yield by accelerating the AD process. The daily methane yield (DMY) of RS after pretreatment with various microbial reagents is shown in Fig. 3. The maximum DMYs of RS pretreated by LC under oxygen concentrations of 0, 6, 12, and 24 mL_{O2}/gVS were 11.5, 22.3, 16.6, and 19.6 mL/gVS on the 3rd day, respectively (Fig. 3A). The maximum DMYs of RS pretreated with CM under different oxygen concentrations were 12.8, 33.0, 21.1, and 18.3 mL/gVS on the 3rd day (Fig. 3B), respectively, whereas the maximum DMYs of RS pretreated with SD for all oxygen concentrations were 13.2, 24, 14.1, and 11.1 mL/gVS between the 3rd and 5th days (Fig. 3C). The maximum DMYs of RS pretreated with BS for various oxygen concentrations were 12.3, 24.7, 16, and 13.5 mL/gVS on the 4th day (Fig. 3D), whereas the maximum DMY for the untreated group was 15.8 mL/gVS recorded on the 6th day, after which methane production declined and became stable from the 16th day onwards. Nevertheless, a different trend was observed in the pretreatment of RS by SC, where the oxygen concentrations of 0, 6, 12, and 24 mL_{O2}/gVS led to significantly higher DMY values of 24.8, 26.6, 31.5, and 24.5 mL/gVS on the 4th day compared with the untreated group, respectively (Fig. 3E). The pretreatment of RS by different microbial reagents, such as LC, CM, SD and BS, showed

higher DMY values at an oxygen concentration of 6 mL_{O2}/gVS, while the SC performed best at an oxygen concentration of 12 mL_{O2}/gVS compared to the URS. These results proved that microbial pretreatment combined with oxygen clearly enhanced methane production efficiency by accelerating the rate of hydrolysis.

3.4.2. Cumulative methane yield (CMY)

The highest CMY values of RS pretreated by the microbial reagents LC, CM, SD, BS, and SC without oxygen were 200.8, 182.0, 193.5, 203.0, and 267.7 mL/gVS, respectively, as shown in Fig. 4. When oxygen was combined with the microbial pretreatment, a further increase in the CMY values was observed. For example, the highest CMYs of RS pretreated with LC, CM, SD, and BS were 257.1, 263.2, 256.7, and 270 mL/gVS at the optimal oxygen concentration of 6 mL_{O2}/gVS, whereas the highest CMY value of RS pretreated with SC was 311.7 mL/gVS at the optimal oxygen concentration of 12 mL_{O2}/gVS. The improvement in the yield by only using LC, CM, SD, BS, and SC was 21.5, 10.2, 17.1, 22.9, and 62.0%, respectively, whereas the applying oxygen led to an increase in the methane yield by 55.6, 59.3, 55.4, 63.7, and 88.7% compared to URS, respectively. An increase in the oxygen concentration beyond the suitable range has a negative effect on the methane yield due to the excess consumption of substrate by facultative organisms under high oxygen levels. Among all reagents, SC outperformed BS, CM, LC, and SD at oxygen concentrations of 12 and 6 mL_{O2}/gVS. Mshandete et al. (2005) reported a 26% higher methane yield from sisal pulp after 9 h of microaerobic pretreatment. Zhu et al. (2009) also reported an increase in the hydrolysis rate during the methane production process by employing a sufficient amount of oxygen. The improvement yield obtained in this study is much higher than previously reported values.

3.5. Structural analysis after microbial pretreatment

3.5.1. SEM

SEM images of untreated and pretreated RS samples were taken at 500× magnification (Supplementary Fig. S1). The URS had a smooth and intact appearance. After microbial pretreatment, the surface of RS pretreated by all microbial reagents without oxygen did not show significant change, whereas the SEM images at 6, 12, and 24 mL_{O2}/gVS oxygen concentrations showed more exhaustive disruption. This change implies that supply with adequate oxygen during microbial pretreatment positively influenced microbial functioning, which eventually led to a greater disruption of RS. Furthermore, microbes exposed to oxygen concentrations of 6 and 12 mL_{O2}/gVS contributed to the highest methane production, implying that these concentrations of oxygen were more suitable for microbial activity. These results demonstrated that microbial pretreatment combined with oxygen could break the compact structure of RS, thereby contributing to the enhanced AD performance.

3.5.2. FTIR

The FTIR spectra of untreated and pretreated RS by five different microbial reagents at oxygen concentrations of 0, 6, 12, and 24 mL_{O2}/gVS are shown in Fig. 5A. The wide band achieved in the range of 3300-2900 cm^{-1} is attributed to levels of intermolecular and intramolecular hydrogen bonds, whereas the peak at 2920.0 cm^{-1} with shoulder bands represents C—H. The absorbance at 1739.2 cm^{-1} is ascribed to stretching of the aromatic ring, which is associated with lignin removal. This absorption peak had a lower intensity in pretreated RS than in URS, showing that a fragment of lignin was disrupted during microbial pretreatment. The prominent wide band at 1242 cm^{-1} is ascribed to C—O in the aromatic ether group based on the presence of acetyl groups in hemicelluloses. The absorbance of this peak disappeared in the RS pretreated by LC, SD, CM, BS, and SC, indicating that the addition of microbial reagents was useful in disrupting hemicellulose. Moreover, the intensity in the 900 cm^{-1} band is very sensitive to the amount of crystalline versus amorphous structure of cellulose. It was seen that the intensity of this band after microbial pretreatment was lower than URS, which indicated higher amount of disrupted structure obtained after pretreatment. The band at 1595 cm^{-1} is attributed to the streching of aromatic ring, which is ascribed to removal of

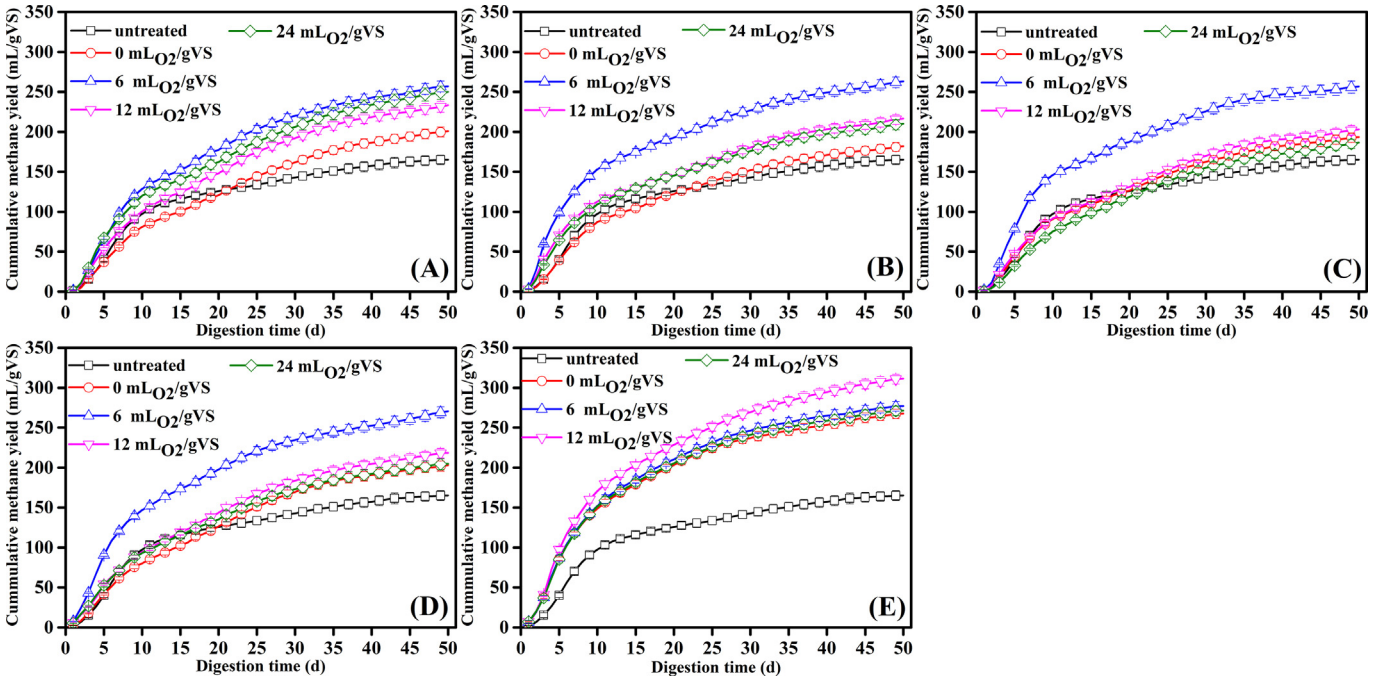


Fig. 4. Cumulative methane yield of RS pretreated by different microbial reagents: (A) LC, (B) CM, (C) SD, (D) BS, and (E) SC at different oxygen concentrations.

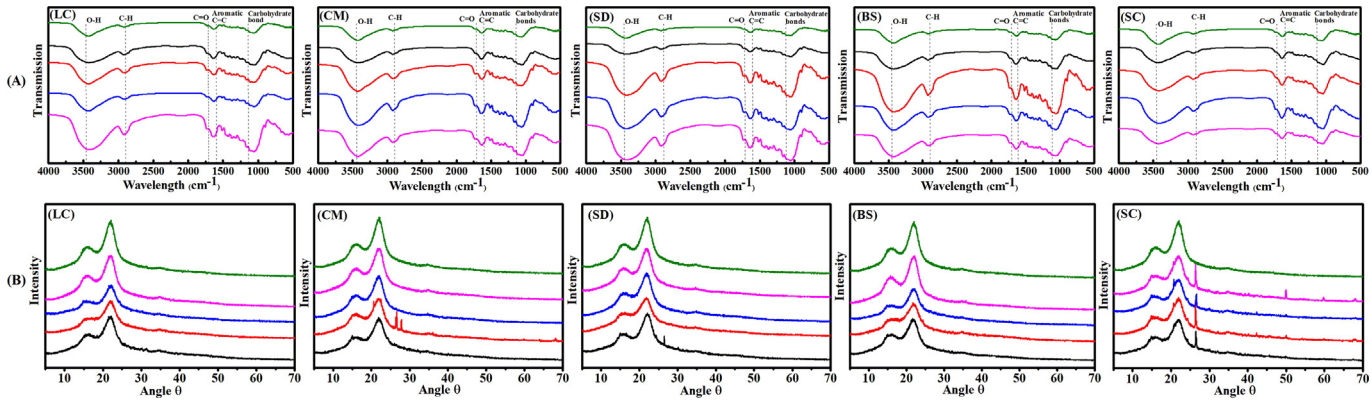


Fig. 5. FTIR (A) and XRD (B) images of RS pretreated by LC, CM, SD, BS, and SC at different oxygen concentrations (untreated, 0 mL_{O2}/gVS, 6 mL_{O2}/gVS, 12 mL_{O2}/gVS, 24 mL_{O2}/gVS).

lignin. These changes were more obvious in the spectra obtained at an oxygen concentration of 6 mL_{O2}/gVS for LC, CM, SD, and BS; while, SC showed the most prominent structural changes at an oxygen concentration of 12 mL_{O2}/gVS. These structural changes in the pretreated RS could be beneficial because lignin removal separated the cellulose fibers, making them more susceptible to microbial degradation. The lower content of lignin in the sample suggested an enhanced efficiency of cellulose hydrolysis, which may lead to improved biodegradability.

3.5.3. XRD

Amorphous cellulose can be easily attacked by different enzymes compared with crystalline cellulose. Microbes attack crystalline cellulose to reduce the crystallinity degree of lignocellulosic material (Zhang et al., 2015). The *CrI* is one of the crucial factors that reflects enzyme hydrolysis during microbial pretreatment. The *CrI* value calculated by Eq. (4) revealed the relative amount of crystalline cellulose. In the current study, the *CrI* of RS was analyzed by XRD, as shown in Table 2. After pretreatment with all microbial reagents at different oxygen concentrations, the crystallinity of RS decreased, indicating that more crystalline cellulose was destroyed, as shown in Fig. 5B. Hence, microbial reagents exposed to suitable oxygen concentrations played a beneficial role in the pretreatment process. Additionally, different microbial reagents performed differently. For example, the highest decrease in the crystallinity index was observed in SD and BS compared to other microbial reagents.

3.5.4. Compositional analysis of lignin, hemicellulose, and cellulose

The changes in the fiber composition of RS pretreated by different microbial reagents at various oxygen concentrations after pretreatment and AD are shown in Table 2. After microbial pretreatment by LC, CM, SD, BS, and SC, the proportion of cellulose increased (34.7 ± 0.3%, 41.8 ± 0.2%, 40.9 ± 0.7%, 40.7 ± 0.5%, and 45.3 ± 0.0%, respectively) compared to the URS (33.9 ± 0.4%), indicating that the lignin content decreased and that cellulose was more accessible to the microbes for breakdown and conversion to the end products of AD. After providing an oxygen supply, a further increase in the proportion of cellulose to 49.6 ± 0.5%, 48.5 ± 0.4%, 49.7 ± 0.3%, 48.4 ± 0.2%, and 47.8 ± 0.4% was achieved at an oxygen concentration of 12 mL_{O2}/gVS for all groups except BS, which showed the highest proportion of cellulose at 6 mL_{O2}/gVS. Microbial pretreatment combined with oxygen played a key role in helping to release not only cellulose but also hemicellulose, as shown in Table 2. After AD, the proportion of cellulose decreased significantly. An increase in lignin content after AD can be ascribed to the degradation of the other components (cellulose and hemicellulose) from RS, thus increasing the relative content of lignin in the digestate (Kong et al., 2018). The results showed that the surface structure of RS was loosen after pretreatment with different microbial reagents in the presence of oxygen and became more accessible to anaerobic bacteria, which promoted the effective hydrolysis of RS.

3.6. Biodegradability (BD)

The BD of RS pretreated with LC, CM, SD, BS, and SC at different oxygen concentrations was calculated and is shown in Table 2. The BD of URS was 38.1%, while microbial pretreatment of RS by LC, CM, SD, BS, and SC without oxygen enhanced the BD to 46.4, 42.1, 44.7, 46.9, and 61.8%, respectively. When the microbial pretreatment was combined with oxygen, a further increase was observed. For example, at 6 mL_{O2}/gVS, the BD of LC, CM, SD, and BS treated group rose to 59.4, 60.8, 59.3, and 62.5%, respectively, whereas the BD of SC treated group was the highest at 72.0% under an oxygen concentration of 12 mL_{O2}/gVS. This finding indicated that the microbial pretreatments facilitated the biodegradation of RS and the combination of the microbial pretreatments with oxygen further enhanced the BD. Therefore, a combination of microbial reagents and a suitable oxygen supply can lead to the desired outcomes in terms of biodegradation and methane production. According to a previous study, the oxygen supply in AD digesters should not be insufficient or excessive because such conditions might inhibit the activity of methanogens and lower the methane yield (Xu et al., 2014). This is supported by the results presented in our study showing that the lowest and highest BD of pretreated RS occurred at oxygen concentrations of 0 and 24 mL_{O2}/gVS and 6 and 12 mL_{O2}/gVS, respectively. Some studies have reported enhanced BD of RS using different pretreatment methods. For example, according to a previous study, the enzymatic pretreatment of RS led to the highest BD of 67.1% on the 30th day of AD (Huang et al., 2020). The BD (%) obtained from this study are higher than those in previously reported literature. The final effluent was characterized after AD. The results of the digestate analysis, including the values of VFAs, TAN, TA, VFAs/TA, and pH, are shown in Table S1. The effluent characteristics indicated that the AD system was stable throughout the experiment.

3.7. Kinetic analysis by the modified Gompertz model

The modified Gompertz model was employed to study the fermentation kinetics of RS, as shown in Table 2. The simulated

Table 2
Digestion performance, parameters of the modified Gompertz model and compositions of RS pretreated by microbial reagents.

Pretreatment microbial reagents	EMY (mL/gVS)	Modified Gompertz Model				BD (%)	Improved yield (%)	CrI (%)	[a]Composition after pretreatment (%)			[a]Composition after AD (%)		
		B_0	μ_m	λ	R^2				Hemicellulose	Cellulose	Lignin	Hemicellulose	Cellulose	Lignin
URS	165.2	155.1	8.1	−0.5	0.969	38.1	–	53.5	–	–	–	25.1 ± 0.5	29.6 ± 0.9	18.8 ± 0.6
LC 0 mL_{O2}/gVS	200.8	205.7	6.1	−1.2	0.987	46.4	21.5	44.3	21.3 ± 0.4	33.7 ± 0.3	29.4 ± 0.7	21.0 ± 0.1	31.5 ± 2.0	31.6 ± 2.2
LC 6 mL_{O2}/gVS	257.1	251.5	9.3	−1.8	0.977	59.4	55.6	42.3	26.7 ± 0.6	44.3 ± 0.3	16.1 ± 1.1	20.8 ± 0.2	32.0 ± 2.0	31.0 ± 2.4
LC 12 mL_{O2}/gVS	233.3	235.5	7.3	−1.9	0.986	53.9	41.2	43.4	27.8 ± 0.3	49.6 ± 0.5	9.5 ± 0.6	22.4 ± 1.0	37.2 ± 1.6	23.9 ± 0.8
LC 24 mL_{O2}/gVS	250.1	251.1	7.7	−3.1	0.975	57.8	51.4	44.7	27.7 ± 0.5	47.3 ± 0.6	8.7 ± 1.1	22.8 ± 0.8	34.1 ± 1.2	25.1 ± 2.6
CM 0 mL_{O2}/gVS	182.0	178.7	6.3	−1.6	0.977	42.1	10.2	48.1	21.8 ± 0.2	41.8 ± 0.2	20.9 ± 0.9	21.4 ± 0.4	39.4 ± 2.3	22.2 ± 1.3
CM 6 mL_{O2}/gVS	263.2	254.0	9.6	−4.2	0.960	60.8	44.6	40.8	28.0 ± 0.3	47.2 ± 0.5	7.7 ± 0.6	21.9 ± 0.1	39.6 ± 0.1	21.3 ± 1.3
CM 12 mL_{O2}/gVS	216.7	216.0	6.5	−4.9	0.967	50.1	31.2	42.1	28.0 ± 0.6	48.5 ± 0.4	10.5 ± 0.3	22.6 ± 0.2	38.9 ± 0.2	19.1 ± 0.5
CM 24 mL_{O2}/gVS	210.1	205.6	7.0	−3.6	0.970	48.5	27.2	44.7	27.1 ± 0.4	48.0 ± 0.2	7.4 ± 1.1	22.2 ± 0.4	36.5 ± 0.5	22.7 ± 2.3
SD 0 mL_{O2}/gVS	193.5	192.9	6.4	−2.0	0.980	44.7	17.1	46.3	21.2 ± 0.4	40.9 ± 0.7	17.8 ± 0.3	21.6 ± 0.3	39.5 ± 1.0	18.3 ± 0.5
SD 6 mL_{O2}/gVS	256.7	249.6	10.1	−2.4	0.964	59.3	55.4	37.7	27.6 ± 0.6	48.4 ± 0.9	7.2 ± 0.4	21.6 ± 0.7	37.1 ± 1.6	20.3 ± 0.6
SD 12 mL_{O2}/gVS	203.1	203.7	6.5	−2.0	0.983	46.9	22.9	39.5	25.9 ± 0.2	49.7 ± 0.3	8.4 ± 0.7	21.3 ± 0.6	38.0 ± 1.1	21.6 ± 0.5
SD 24 mL_{O2}/gVS	186.6	185.1	6.3	−0.6	0.988	43.1	13.0	45.9	27.0 ± 0.5	49.4 ± 1.3	6.3 ± 0.5	21.3 ± 0.2	39.5 ± 1.6	18.1 ± 1.4
BS 0 mL_{O2}/gVS	203.0	208.4	6.4	−1.1	0.991	46.9	22.9	43.4	21.4 ± 0.3	40.7 ± 0.5	20.4 ± 0.9	21.4 ± 0.0	40.5 ± 0.2	19.7 ± 0.6
BS 6 mL_{O2}/gVS	270.5	259.8	10.2	−3.0	0.973	62.5	63.7	37.8	26.9 ± 0.7	48.4 ± 0.2	8.6 ± 0.4	20.6 ± 0.6	38.1 ± 0.1	19.3 ± 0.0
BS 12 mL_{O2}/gVS	218.5	220.5	7.0	−2.0	0.99	50.5	32.3	45.3	27.0 ± 0.8	48.3 ± 0.7	9.2 ± 0.5	20.7 ± 0.0	37.3 ± 0.6	19.8 ± 1.1
BS 24 mL_{O2}/gVS	204.8	206.1	6.6	−2.6	0.987	47.3	24.0	48.2	28.3 ± 0.4	47.0 ± 0.4	9.9 ± 0.8	19.3 ± 0.9	39.7 ± 0.3	23.4 ± 2.1
SC 0 mL_{O2}/gVS	267.7	257.2	11.4	−1.9	0.977	61.8	62.0	45.5	29.3 ± 0.2	45.3 ± 0.0	7.8 ± 0.9	25.4 ± 0.1	28.5 ± 1.6	16.5 ± 1.5
SC 6 mL_{O2}/gVS	277.2	267.9	11.8	−1.8	0.979	64.0	67.8	44.7	30.3 ± 1.3	47.0 ± 0.5	7.5 ± 0.5	24.9 ± 0.3	29.4 ± 1.9	16.8 ± 0.8
SC 12 mL_{O2}/gVS	311.7	300.1	12.1	−2.4	0.971	72.0	88.7	44.5	28.9 ± 0.8	47.8 ± 0.4	7.8 ± 0.9	23.6 ± 1.2	29.5 ± 0.6	17.5 ± 0.0
SC 24 mL_{O2}/gVS	271.3	261.8	11.5	−1.9	0.977	62.7	64.2	47.0	28.7 ± 1.0	47.5 ± 0.3	7.1 ± 1.2	24.3 ± 0.8	28.0 ± 1.3	19.5 ± 0.7

LC: liquid consortium, CM: cow manure, SD: sheep dung, BS: biogas slurry, SC: straw-decomposing consortia, URS: untreated rice straw, EMY: experimental methane yield, B_0: simulated maximum cumulative methane yield (mL/gVS), R^2: correlation coefficient, μ_m: simulated maximum methane production rate (mL/gVS), λ: lag phase time (d), BD: biodegradability, *CrI*: crystallinity index, [a]: composition percentages on a TS basis.

maximum cumulative methane yield (B_0) was close to the EMY, suggesting that the Gompertz model fit well with the CMY curve. This finding was further supported by the high correlation coefficient (R^2), which ranged from 0.96 to 0.99. The time elapsed until a significant production of methane was found during the batch assay is deciphered as the lag phase time (λ). A higher λ means a slower startup, and in this study, all λ values of the pretreated samples were lower than that of the URS, which meant that the RS after microbial pretreatment had a much faster startup process. The maximum methane production rate (μ_m) was positively correlated with the EMY, implying a high AD efficiency. Overall, the parameter analysis demonstrated that the pretreatment of RS with LC, CM, SD, BS, and SC at oxygen concentrations of 6 and 12 mL_{O2}/gVS significantly improved the digestion performance, and among all five microbial reagents, SC showed the highest enhancement of methane production.

3.8. Microbial community analysis of rice straw before and after pretreatment

The microbial analysis of the microbial reagents and RS pretreated by microbial reagents with and without oxygen is displayed in Fig. 6. The results at the phylum level showed that *Firmicutes, Bacteroidetes, Actinobacteria*, and *Proteobacteria* were the most dominant phyla among the microbial reagents, as shown in Fig. 6A. When RS was pretreated with microbial reagents without oxygen, shifts in microbial diversity occurred at the phylum level, and phyla such as *Chloroflexi* and *Spirochaetes* were also detected (Fig. 6B). After providing oxygen, *Firmicutes* became the most abundant phylum in all microbial reagents except LC, followed by *Bacteroidetes* and *Proteobacteria*, whereas *Chloroflexi* and *Spirochaetes* had a relatively low abundance, as shown in Fig. 6C. Hence, the changes in the microbial community structure at the phylum level indicated that microbial pretreatment combined with oxygen increased the abundance of *Firmicutes*, which is known for hydrolyzing carbohydrates and proteins to produce VFAs, indicating that pretreatment can potentially boost the hydrolysis process (Amin et al., 2021; Lim et al., 2014; Lu et al., 2019). Furthermore, a high percentage of bacteria that belong to the phylum *Firmicutes* are facultative and thus show faster growth with oxygen (Fu et al., 2015). *Bacteroidetes* also includes hydrolytic microbes that facilitate the degradation of proteins and complex sugar polymers to VFAs, such as acetate and propionate (Huang et al., 2016; Maspolim et al., 2015), which implied that microaerobic pretreatment caused significant enrichment of hydrolytic bacteria.

Similar trends were seen at the class and order levels. The microbial analysis at the class level showed that *Clostridia, Bacteroidia, Bacillia,* and *Gammaproteobacteria* were the most abundant classes in the microbial reagents, as shown in Fig. 6D. When RS was pretreated with microbial reagents without oxygen, it showed a changing trend in the microbial diversity at the class level, and two more classes *Anaerolineae* and *Flavobacteriia* were also detected, as shown in Fig. 6E. After providing oxygen, *Clostridia* remained the most abundant class in all microbial reagents except LC and BS, followed by *Gammaproteobacteria*, *Bacillia*, and *Bacteroidia*, whereas oxygen had a negative impact on the growth of *Anaerolineae* and *Flavobacteriia*, as shown in Fig. 6F. The microbes from the classes *Clostridia* and *Bacillia* are known to play a key role in the hydrolysis of cellulose as reported in previous studies (Kröber et al., 2009; Wirth et al., 2012). At the order level, *Clostridiales* (CM, SD, BS), *Bacteroidales* (CM, BS), *Enterobacteroidales* (SD), *Lactobacillales* (LC), *Bacillales* (LC, SC), *Pseudomonadales* (SD), *Anaerolineales* (BS), and *Flavobacteriales* (CM) were abundant in the microbial reagents, as shown in Fig. 6G. When RS was pretreated with microbial reagents only, *Clostridiales* and *Bacteroidales* remained

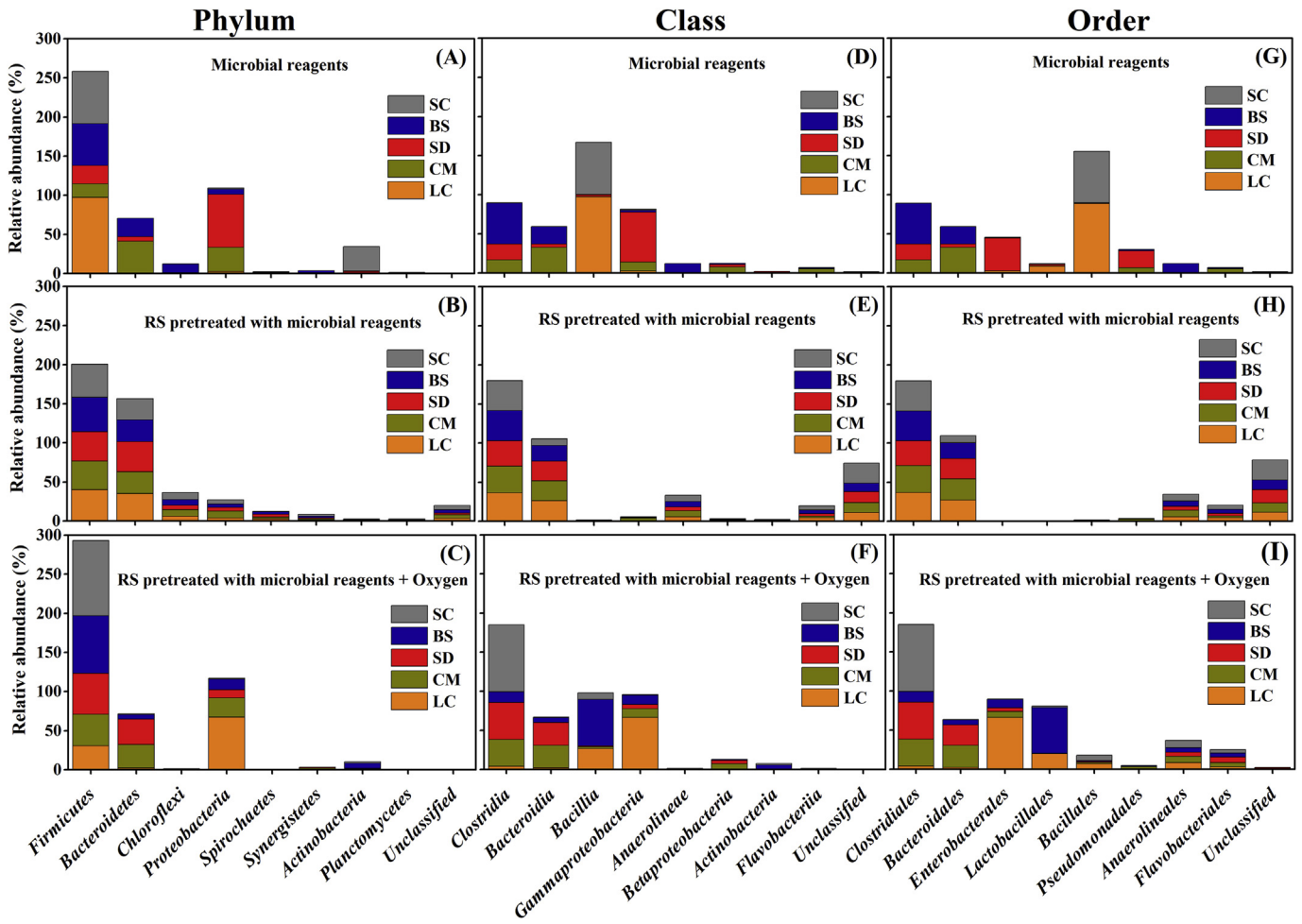


Fig. 6. Taxonomic classifications of the microbial community analysis of microbial reagents, RS pretreated by microbial reagents, and RS pretreated by microbial reagents with oxygen at the phylum (A, B, C), class (D, E, F), and order (G, H, I) levels.

abundant order in all microbial reagents followed by *Anaerolineales* and *Flavobacteriales*, as shown in Fig. 6H. After supplying oxygen, *Clostridiales* continued to be the most abundant order in all microbial reagents except LC and BS, followed by *Enterobacteroidales*, *Bacteroidales*, and *Lactobacillales*, as shown in Fig. 6I. According to previous studies, most members of *Clostridiales* were reported to be able to hydrolyse cellulose (Amin et al., 2021; Wirth et al., 2012).

Thus, after microaerobic pretreatment, the most abundant phyla, classes, and orders were *Firmicutes*, *Clostridia*, and *Clostridiales*, respectively, and they mainly include facultative organisms that are known to play a key role in the hydrolysis of complex polymers to monomers.

3.9. Microbial community diversity after AD

3.9.1. Species diversity and richness

Table 3 shows the richness and diversity estimators, such as the sequence, OTU number, Shannon, ACE, and Chao1 indexes for bacteria and archaea. The ACE and Chao1 indexes indicate richness, whereas the Shannon index indicates the diversity of microbes (Zhi et al., 2019). The OTU number, ACE index, and Chao1 index for bacteria and archaea were lower in the AD digestate of RS pretreated by LC, CM, SD, and BS compared to URS. Variations in the OTU number and ACE and Chao1 index were observed in the groups supplied with oxygen compared with the groups without oxygen. For example, regarding bacteria, oxygen positively influenced these indicators in SD, BS, and SC, whereas in archaea, oxygen increased these indicators in all microbial groups. For SC, these indicators were even higher than those of the URS, indicating microbial enrichment and diversification, which might have facilitated the hydrolysis and conversion rate and eventually resulted in the highest methane production by this microbial reagent. The sequencing depth for all samples was sufficient to cover the whole diversity, which was confirmed by the coverage values of 98–99% for bacteria and archaea. The diversity of microorganisms is also presented in Fig. S2.

3.9.2. Bacterial genera

An analysis of the microbial community after AD of RS pretreated with five microbial reagents without and with oxygen is shown in Fig. 7A and B. The most abundant genera were *Clostridium III*, *Saccharofermentans*, *Clostridium sensu stricto*, *Levilinea*, *Sedimentibacter*, *Lutaonella*, *Proteiniphilum*, *Syntrophomonas*, *Sphaerochaeta*, *Candidatus Cloacamonas*, *Smithella*, *Petrimonas*, *Prolixibacter*, *Meniscus*, and *Mariniphaga*, which belong to the phyla *Firmicutes*, *Bacteroidetes*, *Proteobacteria*, *Chloroflexi*, and *Spirochaetes*. It was observed that different bacteria responded differently to the oxygen supply, and some bacterial genera were positively influenced by the oxygen supply, whereas some of them had high RA without oxygen. In addition, each microbial reagent showed a varying response to the oxygen supply. The RA of *Syntrophomonas* and *Candidatus Cloacamonas* increased with oxygen in all groups, whereas that of *Levilinea* remained almost the same with

Table 3
Indexes indicating the microbial diversity and richness in digestate of pretreated RS by different microbial reagents.

	Sample ID	Seq num	OTU	Shannon	ACE	Chao 1	Coverage
Bacteria	URS	64098	3451	4.8	48393.6	22008.2	0.98
	LC 0 mL_{O2}/gVS	63974	2026	4.9	4537.7	3385.0	0.99
	LC 6 mL_{O2}/gVS	47914	1980	5.2	3966.6	3124.5	0.99
	CM 0 mL_{O2}/gVS	48904	2022	5.2	3912.5	3071.1	0.98
	CM 6 mL_{O2}/gVS	34876	1738	5.1	3665.4	2801.8	0.98
	SD 0 mL_{O2}/gVS	44061	1833	5.0	3817.6	2983.0	0.98
	SD 6 mL_{O2}/gVS	48839	2160	5.1	4891.5	3606.2	0.98
	BS 0 mL_{O2}/gVS	43807	1844	5.2	4033.9	3040.3	0.98
	BS 6 mL_{O2}/gVS	48120	1916	5.0	3764.4	3072.1	0.98
	SC 0 mL_{O2}/gVS	53438	3674	5.1	53569.4	24818.9	0.98
	SC 12 mL_{O2}/gVS	69716	3991	4.9	60000.4	28531.6	0.98
Archaea	URS	98722	2360	1.7	66882.4	27794.1	0.98
	LC 0 mL_{O2}/gVS	67160	1072	2.7	2486.4	1775.0	0.99
	LC 6 mL_{O2}/gVS	70205	1157	2.5	3471.4	2297.1	0.99
	CM 0 mL_{O2}/gVS	73868	1243	2.7	3027.7	2169.8	0.99
	CM 6 mL_{O2}/gVS	71173	1197	2.5	2945.3	2072.8	0.99
	SD 0 mL_{O2}/gVS	65825	1176	2.5	3141.1	2097.2	0.99
	SD 6 mL_{O2}/gVS	68918	1319	2.6	3563.3	2375.1	0.99
	BS 0 mL_{O2}/gVS	65357	1138	2.4	2773.1	2021.8	0.99
	BS 6 mL_{O2}/gVS	67502	1196	2.8	2865.4	2074.2	0.99
	SC 0 mL_{O2}/gVS	60194	2391	1.8	108326.1	34597.9	0.98
	SC 12 mL_{O2}/gVS	62273	2675	2.0	111094.8	38300.1	0.98

URS: untreated rice straw, LC: liquid consortium, CM: cow manure, SD: sheep dung, BS: biogas slurry, SC: straw-decomposing consortia.

and without oxygen. The RA of *Clostridium III* and *Lutaonella* increased with oxygen in groups including CM, SD, and BS, whereas LC had a high RA without oxygen. Similarly, the RA of *Saccharofermentans* and *Sedimentibacter* increased with oxygen in all groups except CM, which had a higher growth without oxygen. The RA of *Clostridium sensu stricto* decreased with oxygen in all groups, and the RA of *Sphaerochaeta* also decreased with oxygen in all groups except CM, whereas in SC it was not detected. This indicated that oxygen had a negative impact on the growth of *Clostridium Sensu stricto* and *Sphaerochaeta*. Oxygen also had a negative impact on the RA of *Prolixibacter*, *Petrimonas*, and *Mariniphaga* in all groups except LC and SC. The RA of *Proteiniphilum* decreased in all groups with oxygen except BS and SC. Overall, microbial pretreatment combined with oxygen seemed to enhance the microbial diversity compared to the URS. It was interesting to note that the genus *Smithella* and *Lutaonella* were hardly detected in the URS but were found in the AD digesters after microbial pretreatment in all groups except SC. The genus *Smithella* increased in all groups with oxygen, whereas *Lutaonella* increased significantly in CM, SD, and BS with oxygen but decreased in LC.

Clostridium III had a very low RA in the URS digestate (1.6%), although its RA increased from 2.6 to 11.3% with and without oxygen in all groups. These bacteria follow hydrolytic pathways and degrade cellulose to produce VFAs (Horino et al., 2014; Li et al., 2018b). The RA of *Saccharofermentans* was 5.5% in the URS digestate and increased to 5.9%–12.4% after microbial pretreatment with and without oxygen in all groups. These bacteria follow acidogenic pathways and utilize sugars to produce acetate, lactate, and fumarate (Chen et al., 2010; Hahnke et al., 2016). *Clostridium sensu stricto* had the highest RA in the URS digestate (16.9%), which decreased to 1.0–8.2% with and without oxygen in all groups. These bacteria follow both hydrolytic and acidogenic pathways to utilize many sugars as carbon and energy sources for producing methanogenesis precursors such as acetate, butyrate, H_2, and CO_2 (Li et al., 2018a). The RA of *Levilinea,* which mainly converts sugars into acetate, lactic acid, and H_2, did not show a significant difference in all pretreated groups (6.7–12.1%) compared to the URS group (13.2%). *Sedimentibacter* had the highest RA of 18.0% and 13.6% in the RS digestate pretreated with SC with and without oxygen supply, respectively, whereas it showed a lower RA in the RS digestate pretreated with other microbial reagents and URS. These bacteria mainly use amino acids as fermentation substrates to produce VFAs (Imachi et al., 2016; Li et al., 2018a). *Lutaonella* follows hydrolytic pathways and had a high RA in the RS digestate pretreated by all the microbial reagents (0.1–9.2%) with and without oxygen compared to the URS (0.03%), and these bacteria facilitate polysaccharide fermentation and produce VFAs and H_2 that are vital for methanogenesis (Arun et al., 2009). The RA of *Proteiniphilum* and *Syntrophomonas* in the URS digestate was low (2.8% and 2.0%) compared to the digestate of RS pretreated by different microbial reagents with and without oxygen (3.8–7.6% and 3.0–4.6%). *Proteiniphilium* utilizes sugars, whereas *Syntrophomonas* is a hydrogen-producing acetogenic genus that works in syntrophic association with methanogens to convert H_2 and VFAs, which are further converted into methane (McInerney et al., 1981). *Sphaerochaeta* had a higher RA in all pretreated groups with and without oxygen (0.05–5.3%) than URS (0.03%). These bacteria are chemoheterotrophic anaerobes that follow hydrolytic pathways to ferment carbohydrates to produce acetate, formate, and ethanol (Ritalahti et al., 2012). The RA of *Candidatus Cloacamonas* was lower in all pretreated groups without oxygen (0.7–4.4%), compared to the groups pretreated with LC (7.28%), CM (6.6%), and SC (6.1%) with oxygen compared to the URS (4.7%). This bacterium is able to produce hydrogen *via* fermentation and its iron-hydrogenases, making it possible to syntrophically associating with methanogens (Pelletier et al., 2008). *Smithella* had the highest RA in the digestate of RS pretreated with LC, CM, SD, and BS with and without oxygen (1.7–3.6%), whereas it was not found in SC and URS. This genus is commonly found in anaerobic digesters and works in syntrophic association with methanogens, specifically *Methanosaeta,* during the methane production process (Amin et al., 2021; Zamorano-López et al., 2020). The RA of *Petrimonas* did not show a significant difference in the pretreated groups (1.0–2.5%) compared to the URS group (2.1%). These bacteria follow both hydrolytic and acidogenic pathways to ferment carbohydrates and some organic acids to acetate, propionate, CO_2, and H_2 (Huang et al., 2016; Maspolim et al., 2015). *Candidatus Hydrogenedens* had the highest RA of 0.01-3.6% in the digestate of all pretreated groups with and without oxygen compared to the URS (0.01%), and these bacteria are responsible for carbon fixation and hydrolyze lipids, long-chain

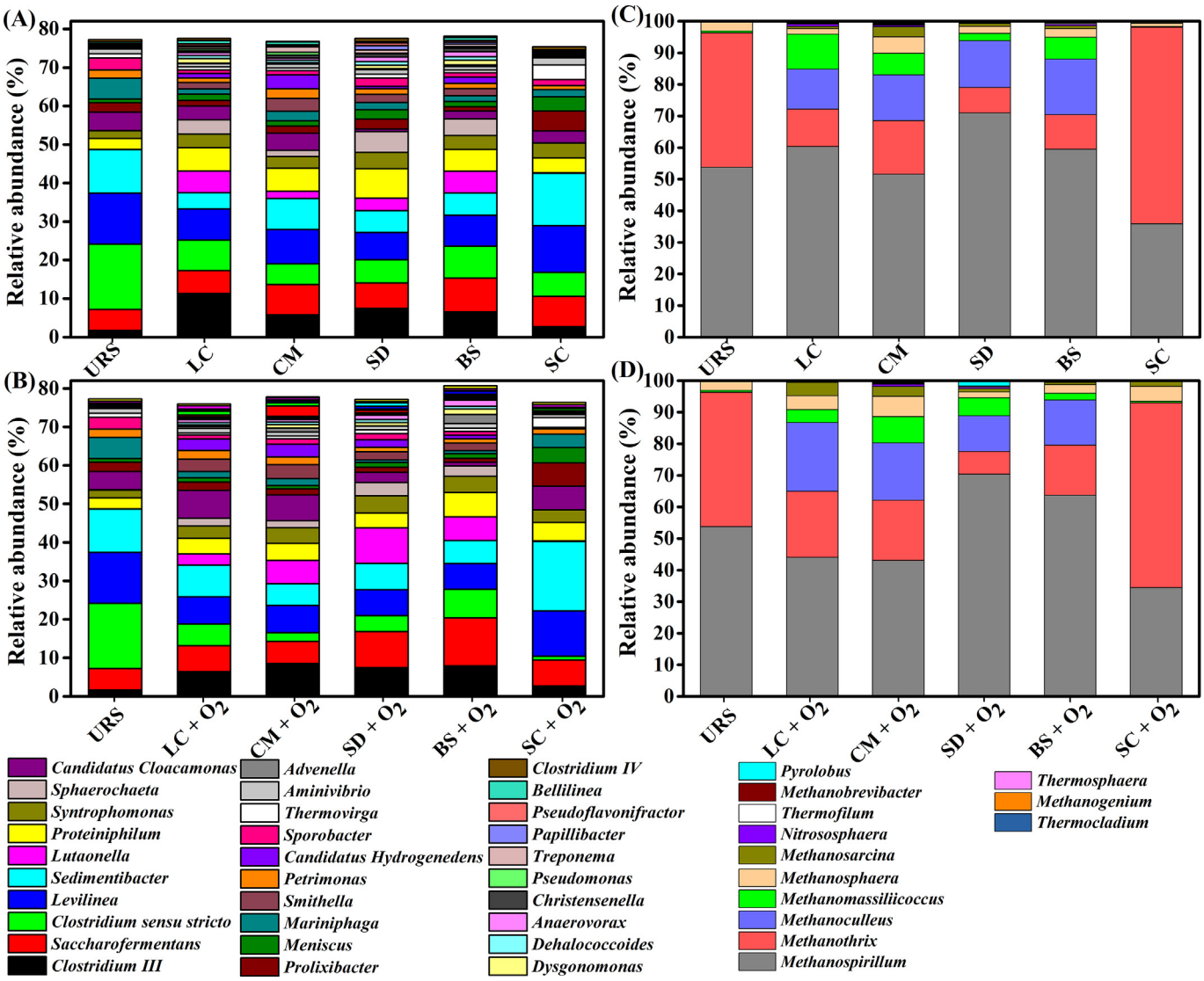


Fig. 7. Relative abundance of dominant bacterial (A and B) and archaeal (C and D) genera of RS pretreated with the microbial reagents LC, CM, SD, BS, and SC with and without oxygen.

fatty acids, and triglycerols to produce CO_2 and acetate (Momper et al., 2018). The RA of *Proxilibacter* was lower in the digestates of RS pretreated by LC, CM, SD, and BS both with and without oxygen (1.0–2.3%) compared to the URS (2.4%) but higher in RS pretreated by SC (6.0%), whereas the RA of *Mariniphaga* was highest in the URS compared to other pretreated groups with and without oxygen. *Proxilibacter* and *Mariniphaga* are facultatively aerobic and utilize sugars to produce fatty acids (Iino et al., 2014).

Based on the microbial analysis, *Clostridium III*, *Syntrophomonas*, *Candidatus Cloacamonas*, *Levilinea*, *Lutaonella*, *Saccharofermentans*, and *Sedimentibacter* presented the highest contributions to methane production during AD of RS by hydrolyzing carbohydrates, which eventually led to improved methane yield. Furthermore, a large percentage of bacteria affiliated with the phylum *Firmicutes* are facultative or aerobic bacteria, which have a good growth rate with oxygen, indicating that oxygen supply in combination with microbial pretreatment enrich the useful microbes to carry out the AD process (Fu et al., 2015).

3.9.3. Archaeal genera

The composition of the methanogens in the RS digestate pretreated with LC, CM, SD, BS, and SC with and without oxygen at the genus level was further investigated, as shown in Fig. 7C and D. The major genera of methanogens were *Methanospirillum*, *Methanothrix, Methanoculleus*, *Methanomassiliicoccus, Methanosphaera,* and *Methanosarcina,* which belong to the phylum *Euryarchaeota*. All groups had different microbial structures, indicating that differences in microbial reagents finally influenced the microbial community dynamics during the AD process. It was interesting to note that in the URS and SC digestates, *Methanospirillum* and *Methanothrix* were predominant but presented different ratios. For example, in the URS, hydrogenotrophic *Methanospirillum* was the predominant archaeon and had a RA of 53.7% while acetoclastic *Methanothrix* had an RA of 42.4%; however, after microbial pretreatment with SC (with and without oxygen), the RA of acetoclastic *Methanothrix* increased to the range of 58.5–62.1%, suggesting that the methanogen type changed from hydrogenotrophic to acetoclastic. Hydrogenotrophic *Methanospirillum* was the highest in SD and then BS, which had RA values of 70.3–71.0% and 59.4–63.6%, respectively, with and without oxygen supply. *Methanospirillum* mainly uses formate, formic acid or H_2/CO_2 to produce methane (Iino et al., 2010; Zhou et al., 2014), whereas *Methanothrix* utilizes acetate exclusively

to produce methane and CO_2 (Huser et al., 1982). Archaea such as *Methanoculleus* and *Methanomassiliicoccus* were almost nonexistent in the SC digestate, whereas these archaea had a high RA in the LC, CM, SD, and BS digestate with and without oxygen supply. This finding suggested that digestates of LC, CM, SD, and BS contained hydrogenotrophic and methylotrophic methanogens while that of SC mainly contained acetoclastic and hydrogenotrophic methanogens, which might contribute to the enhanced methane production by this group. *Methanoculleus* utilizes a wide range of substrates for methanogenesis, such as H_2/CO_2 and formate (Chen et al., 2015; Shimizu et al., 2013), whereas *Methanomassiliicoccus* uses H_2 as the electron donor and produces methane by reducing methanol (Dridi et al., 2012). *Methanosarcina* had higher RA in LC (4.3%) and CM (3.1%) with oxygen compared to that without oxygen and the URS. Members of the genus *Methanosarcina* are particularly attractive because species from this genus are capable of growing on several simple carbon compounds using any of the three methanogenic pathways: acetoclastic, hydrogenotrophic, and methylotrophic (Archer and King, 1983; Wang et al., 2018). In contrast, most other known methanogen species are restricted to using CO_2 and H_2 for hydrogenotrophic, acetate for acetoclastic, and methanol for methylotrophic growth. Therefore, their metabolic flexibility and occurrence in the digestates of LC and CM are advantageous in the methane production process. Hydrogen-dependent methylotrophic methanogen *Methanosphaera* was also detected in all groups, and it produces methane from methanol and H_2 only, while interestingly, 95% of the cell carbon is synthesized from acetate and CO_2, which are not used for methane production (Miller and Wolin, 1985; Weiss et al., 2008). The RA of this archaeon was 2.8% in the URS but ranged from 1.0% to 6.4% in the pretreated groups.

Compared with the URS, higher RA values were observed for *Methanospirillum*, *Methanoculleus*, and *Methanomassiliicoccus* in LC, SD, and BS without oxygen; *Methanoculleus*, *Methanomassiliicoccus*, *Methanosphaera*, and *Methanosarcina* in CM without oxygen; and *Methanothrix* in SC without oxygen. However, compared with the URS, enhanced RA values were observed of *Methanoculleus*, *Methanosphaera*, *Methanoassiliicoccus*, and *Methanosarcina* in LC and CM with oxygen; *Methanospirillum*, *Methanomassiliicoccus, Methanoculleus*, and *Methanosarcina* in SD with oxygen; *Methanospirillum*, *Methanoculleus*, and *Methanoassiliicoccus* in BS with oxygen; and *Methanothrix*, *Methanosphaera,* and *Methanosarcina* in SC with oxygen. These findings indicated that changes in methanogen types, such as acetoclastic, hydrogenotrophic, and methylotrophic methanogens, were caused by precursor changes resulted from the pretreatments and different microbial metabolisms, which impacted the methane production process by different microbial reagents.

4. Energy potential analysis and practical implications

According to the conclusions from the methane yield, an energy potential analysis was carried out to compare the AD performance of different microbial reagents used to enhance the energy output of RS.

4.1. Energy potential analysis

The energy potential analysis of the RS after microaerobic pretreatment was calculated. Among all the microbial reagents used for pretreatment, the SC pretreatment of RS showed the highest energy density (8.67×10^3 kJ kg^{-1}), followed by BS (7.52×10^3 kJ kg^{-1}), CM (7.32×10^3 kJ kg^{-1}), LC (7.15×10^3 kJ kg^{-1}), and SD (7.14×10^3 kJ kg^{-1}), whereas the energy density of URS was 4.59×10^3 kJ kg^{-1}. In China, the yield of RS was estimated to be 213 million tons (Xia et al., 2018); if this amount of RS is converted to bioenergy *via* AD after microbial pretreatment, the overall energy yield is estimated to be in the range of 1.52×10^{12}-1.85×10^{12} MJ/year using different microbial reagents. This finding indicated that there is a huge energy potential for implementing microbial pretreatment in AD of RS.

4.2. Practical implication

The microbial pretreatment technique facilitates the bioconversion of organic matter to biomethane by providing a practical pretreatment method for lignocellulosic material through utilizing readily available microbial reagents along with an oxygen supply to enhance the methane yield. Additionally, the residual digestate from AD can be used as an organic fertilizer, thereby promoting nutrient recovery and crop production.

5. Conclusions

Microaerobic pretreatment represents an effective pretreatment method for the biodegradation of RS with enhanced methane production during AD. All five microbial reagents, namely, SC, BS, CM, LC, and SD, represented sources of useful bacteria and led to an increase in methane yield by 88.7%, 63.7%, 59.3%, 55.6%, and 55.4%, respectively. The most suitable oxygen concentrations were 6 and 12 mL_{O2}/gVS and increases in oxygen concentration beyond these values led to a decrease in the methane yield. The results of the microbial community analysis showed that the use of microbial reagents improved the microbial diversity in the AD digester. When microbial reagents with varying oxygen concentrations were used, the microbial population was further enriched, which eventually led to enhanced methane production performance. The microbial community analysis revealed that the RA of *Firmicutes* increased after microaerobic pretreatment, which is known to play a key role in accelerating the rate of hydrolysis. The RA of *Clostridium III*, which facilitates cell wall degradation by following hydrolytic pathways, along with that of other sugar-fermenting bacteria, such as *Saccharofermentans*, *Lutaonella*, and *Sedimentibacter*, increased after AD in the digestate of pretreated RS compared to the URS. The archaeal richness of *Methanospirillum, Methanothrix, Methanoculleus*, and *Methanomassiliicoccus* also changed after AD in the pretreated RS compared to the URS. This result indicated that pretreatments led to changes in microbial metabolism, which eventually caused a shift in methanogen pathways, such as acetoclastic, hydrogenotrophic, and methylotrophic pathways, and therefore impacted the methane production process. SC was found to be the most efficient microbial reagent for the degradation of RS compared to BS, CM, LC, and SD. The energy potential analysis showed that the energy yield increased after pretreatment with all groups (1.52×10^{12}-1.85×10^{12} MJ/year) compared with the energy yield of the URS (0.98×10^{12} MJ/year). This finding implies that microbial pretreatment is a suitable technique for enhancing the energy conversion efficiency of RS and a promising pretreatment strategy for the treatment of other lignocellulosic materials for their application in AD at the industrial scale.

CRediT authorship contribution statement

Farrukh Raza Amin: Methodology, Conducting experiments, Original draft preparation. **Habiba Khalid:** Data processing & draft preparation. **Wanwu Li:** Effluent measurement. **Chang Chen:** Designing, Supervision, Reviewing and Editing. **Guangqing Liu:** Supervision and Reviewing.

Declaration of competing interest

The authors declare that they have no known competing

financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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